

The Shadow Price of Carbon in Economic Analysis

2024 Guidance Note

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Abbreviations

CBA	Cost-Benefit Analysis
CCAP	Climate Change Action Plan
CEA	Cost-Effectiveness Analysis
CO ₂ e	Carbon Dioxide Equivalent
CPI	Consumer Price Index
ERR	Economic Rate of Return
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GP	Global Practice
IBRD	International Bank for Reconstruction and Development
IDA	International Development Assistance
IFI	International Financial Institutions
LCA	Least-Cost Analysis
NPV	Net Present Value
OP/BP	Operational Policy / Bank Procedure
OPCS	Operational Practices & Country Services
O&M	Operation and Maintenance
PAD	Project Appraisal Document
PV	Present Value
SPC	Shadow Price of Carbon
tCO ₂ e	Tons of Carbon Dioxide Equivalent
USD	United States dollars
WB	World Bank

This guidance note is intended to help World Bank (WB) staff value carbon emissions and emission reductions in the economic analysis of investment project financing. The economic analysis is requested under the Operational Policy and Bank Procedure (OP/BP) 10.00. The guidance provided in this note aims to enhance the economic analysis by using the shadow price of carbon for applicable projects. It replaces the 2014 “Social Value of Carbon in Project Appraisal Guidance Note” and the 2017 “Guidance note on shadow price of carbon in economic analysis.” The note will be updated and complemented periodically, based on new knowledge and feedback from teams.

1. Applicability

The use of shadow prices of carbon (SPCs) in the economic analysis is a corporate commitment¹ for all investment project financing by the International Bank for Reconstruction and Development (IBRD) and the International Development Assistance (IDA) that is subject to greenhouse gas (GHG) emissions accounting. GHG emissions accounting is undertaken for IBRD and IDA investment finance projects in the Global Practices (GPs) with WB-approved GHG accounting methodologies.² Projects that are not subject to GHG emissions accounting do not have to use SPCs in the economic analysis.³ The corporate requirement to apply SPCs in economic analysis went into effect on July 1, 2017, applying to projects with concept notes approved on or after that date. The rest of this note is confined to those project activities for which WB-approved GHG accounting methodologies are available and which are subject to GHG emissions accounting⁴.

2. Background

In 2015, the world came together and agreed to limit global warming to well below 2°C above pre-industrial levels and make efforts to limit warming to 1.5°C in the Paris Agreement. In the Climate Change Action Plan (CCAP) 2016–2020 and the updated CCAP 2021–2025, the WB committed to supporting the achievement of these goals with increased commitments to climate finance and delivery of projects with avoided GHG emissions.

¹ See [Climate Change Action Plan 2021–2025](#) and [IDA 19 commitments](#). Carbon prices used in economic analysis have been referred to as shadow prices of carbon, social costs of carbon, and social values of carbon. These terms refer to different approaches to capturing the price of carbon emissions. This guidance note uses the term “shadow price of carbon,” which is the price of carbon consistent with a given climate mitigation objective.

² Available at the WB’s internal portal [GHGACCOUNTING/](#) under Sectoral Resources.

³ An example is a project with emissions that do not meet GP-specific minimum thresholds for GHG emissions accounting.

⁴ In accordance with the World Bank Paris Alignment methods for IPF and PforR, the SPC can also be used as one of the tools to inform the demonstration of the Paris alignment (PA) of Bank’s operations. The World Bank Paris Alignment Sector Notes may provide sector-specific guidance on applying SPC as part of the PA assessments, which should be used in conjunction with this guidance note. Regarding the Energy and Extractives PA Sector Note: When project economic analysis uses least-cost decarbonization modeling, the exogenous carbon prices discussed in this Guidance Note are not applicable. This is because least-cost decarbonization modeling yields endogenous carbon prices, rendering the two methodologies incompatible. Where relevant, the Bank will be developing a supplementary guidance on applying the SPC as a tool to inform PA assessment and providing it in due course.

Projects financed by the WB can have an impact on GHG emissions. GHG emissions are global externalities with wide-ranging socioeconomic and environmental impacts. Different approaches have been used to internalize these carbon externalities in economic analysis to inform policy and investment decisions:

- The **social cost of carbon** (SCC) is an estimate of monetized damages of emitting one additional unit of carbon dioxide equivalent (CO₂e) into the atmosphere. Although intended to be a comprehensive metric that captures the marginal global damages from a unit of carbon emissions in the cost-benefit analysis (CBA) of policy decisions, current limitations of modeling to estimate SCCs result in large uncertainties about the true marginal damage costs of GHG emissions. These uncertainties exist because many vitally important risks and costs (such as the collapse of the Greenland ice sheet and material changes in ecosystems) associated with climate change are difficult to capture in the SCC, and because the SCC is highly dependent on risk aversion and judgment values regarding non-market impacts and intra- and inter-generational inequality, which have been discussed extensively in the literature (IPCC 2014a,b; Tol 2012; Stern 2013; Weitzman 2014; Dietz and Stern 2015). Because of its dependency on values and worldviews, the SCC is and will always remain contested.
- The **shadow price of carbon** (SPC) avoids some issues in estimating SCCs because its calculation is based on achieving the objective of limiting the global temperature rise and reducing global GHG emissions correspondingly. For a global objective such as limiting global warming below 2°C, the SPC represents the price of carbon that—if implemented fully and on time across the global economy—would ensure the achievement of the objective.⁵ There are uncertainties associated with the estimation of SPCs, such as economically viable technology options in the future and their associated costs, future fuel prices, and policy efficiency, even if these uncertainties are arguably smaller than those for the SCC. If the objectives of the Paris Agreement are considered “optimal”, the SCC and the SPC are equal.

This guidance note provides a corporate approach to incorporating carbon externalities in the scenarios considered in the economic analysis.⁶ One objective is to support the implementation of the Paris Agreement and the achievement of its objectives. To that end, **SPC trajectories representing the path for carbon emissions that are consistent with achieving the temperature objective of the Paris Agreement have been selected.** In practical terms, it is recommended to use the SPC values proposed by the High-Level Commission on Carbon Prices, led by Joseph Stiglitz and Nicholas Stern. The prices are extrapolated beyond the 2030-time horizon⁷ and deemed consistent with achieving the temperature goal of the Paris Agreement. This is presented in detail under section 3 below.

⁵ In the optimization framework, the SPC is also a measure of the change in social welfare resulting from a marginal change in the objective, expressed in terms of carbon emissions, which is why it is also sometimes called the social value of carbon. The Nobel-prized economist Jan Tinbergen defines SPC as “prices indicating the intrinsic or true value of a factor or product in the sense of equilibrium prices”.

⁶ Financial analysis considers actual monetary flows and includes taxes and subsidies not captured in economic analysis, and excludes the social value of carbon (but includes any carbon tax or similar instruments that require project stakeholders to pay a fee for their emissions) and other externalities not paid for.

⁷ The Commission focused on explicit carbon pricing and assumed a supportive policy environment and the parallel implementation of policies to tackle the other externalities impairing emissions reduction, such as economies of scale, or knowledge spillover slowing down innovation. For this reason, the value proposed by the Commissions are lower than other proposed values in the literature.

As detailed in the [Investment Project Finance Economic Analysis Guidance Note](#), the economic analysis for WB investment project finance generally falls under one of two general categories: (1) cost benefit analysis (CBA), and (2) cost-effectiveness analysis (CEA) or least-cost analysis (LCA):

- (1) The **cost-benefit analysis** monetizes all major benefits and costs associated with the proposed project so that they can be directly compared with each other as well as to reasonable alternatives to the project. CBA is generally considered the most comprehensive approach. The calculations include social and environmental costs to society that are not captured in financial analysis, which is seen from the point of view of the project-implementing entity that depends on cost recovery for sustainability. GHG emissions with SPCs are costs. To assess whether benefits outweigh costs, the net present value (NPV) of the project, taking the difference between the present values (PVs) of the benefits and costs, is usually used. A project is considered economically beneficial if its NPV is positive and is higher than other options being examined.
- (2) The **cost-effectiveness** or **least-cost analysis** determines which of several project options achieves a given objective at the lowest cost. Cost-effectiveness may be expressed in dollars discounted costs per unit of output or per unit of benefit and least cost as a comparison of the present values of all costs for the project and its alternatives. Where the project and all alternatives deliver the same level of service (service hereinafter)⁸ within the same timeframe,⁹ the benefits related to the direct provision of service are identical, even though other types of benefits may not be. CEA and LCA are similar to CBA in many important respects but do not attempt to monetize all anticipated benefits from the project or the alternatives considered. The emissions generated by a project can be considered as a cost for the project (and this cost would be negative if the emissions are negative).

The economic analysis of projects is one of the many considerations that inform decision-making by the WB and its clients on whether or not to undertake a project. As such, the purpose of this note is to contribute to greater transparency and consistency of the project's global climate impacts to help decision-making by the WB and its clients. This note is not intended to provide a criterion for project selection; rather, it provides a methodology to ensure that the management and the Board have the information needed to make decisions.

It is possible that an emission-increasing project (e.g., new transport infrastructure) is economically beneficial from a purely domestic point of view but is not economically viable when the global (negative) carbon externality is accounted for. An economic analysis that incorporates carbon externality can encourage WB teams and their clients to consider alternative project implementation arrangements with the use of benefit reallocation mechanisms, notably financial mechanisms such as concessional finance or sales of carbon credits, that allow countries to capture part of the global benefits of avoided carbon emissions.

The SPC and its impact on the economic analysis is conceptually distinct from the question of how the project should be financed or who should pay for the global externality cost or gain from the global

⁸ Projects in this guidance note are understood to be delivering goods, services, or both. For brevity, goods, services, or both are abbreviated as service in the rest of the document.

⁹ Any differences in the level of service delivered need to be quantified to ensure comparability among options. Such quantification may be necessary where the options considered deliver different levels of service and yet the full information to perform CBA quantifying all benefits is not available. This illustration points to the boundary between CBA and LCA not being sharply drawn in practice.

externality benefit of the project. The financing questions are not covered in this guidance note, which focuses only on the use of SPCs in economic analysis. Annex 1 of this guidance note summarizes the recent developments that have led to the revision of the 2014 and 2017 SPC guidance notes.

3. Valuation of GHG emissions

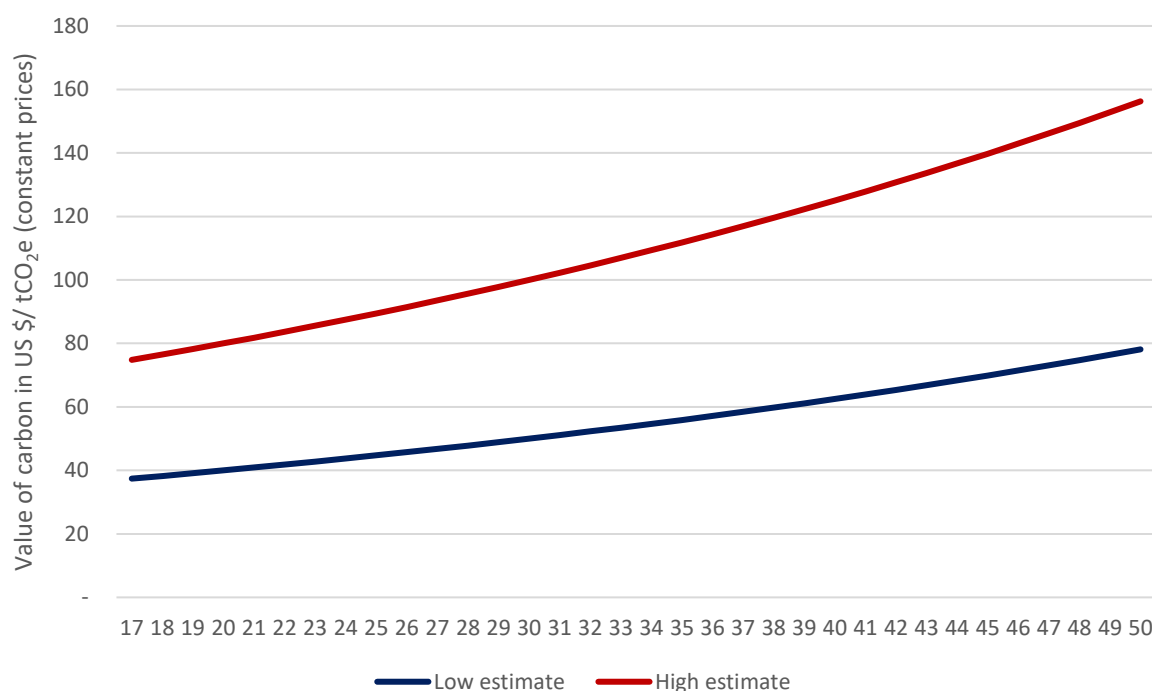
The [High-Level Commission on Carbon Prices](#), led by Joseph Stiglitz and Nicholas Stern, carried out an extensive review of the literature and concluded that carbon prices in the range of USD 40–80 per ton of CO₂e in 2020, rising to USD 50–100 per ton of CO₂e by 2030, is consistent with achieving the temperature goal of the Paris Agreement, provided a supportive policy environment is in place.¹⁰

It is important to note that these values were estimated assuming a supportive policy environment, including appropriate regulations, public investments, and functioning capital markets. As such, the social value of carbon is a useful but insufficient criterion to determine whether a project is consistent with the long-term decarbonization of the world economy, which is required to keep the increase in global temperature below 2°C.

In line with the High-Level Commission on Carbon Prices, this guidance note recommends that project economic analysis use a low and a high estimate of the carbon price, i.e., starting at USD 40 and USD 80, respectively, in 2020 and increasing to USD 50 and USD 100 by 2030, all expressed in 2017 USD equivalent. Given that the *Report of the High-Level Commission on Carbon Prices* does not prescribe any specific carbon price values beyond 2030, the low and high values on carbon prices are extrapolated from 2030 to 2050 using the average annual growth rate of 2.25% per year between 2020 and 2030, leading to values of USD 78 and USD 156 for low and high values, respectively, by 2050.¹¹ Figure 1 shows the low and high values of SPC that all the projects are recommended to use for the period 2017–2050.

¹⁰ Carbon Pricing Leadership Coalition 2017. Report of the High-Level Commission on Carbon Pricing, Commission chairs: Stiglitz, J.E. and Stern, N., supported by World Bank Group, ADEME, French Ministry for the Ecological and Inclusive Transition. https://static1.squarespace.com/static/54ff9c5ce4b0a53decccfb4c/t/59244eed17bffc0ac256cf16/1495551740633/CarbonPricing_Final_May29.pdf. Based on the date of publication, the prices cited in the report are assumed to be expressed in 2017 USD.

¹¹ This is a conservative assumption reflecting a very optimistic forecast of early mitigation action and rapid cost decline of low-carbon technologies

Figure 1. Recommended SPC in USD per metric ton of CO₂ equivalent (constant prices in 2017 USD)

Year	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
Low	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	55	56	57	58	60	61	63	64	65	67	68	70	71	73	75	76	78
High	75	77	78	80	82	84	86	87	89	91	94	96	98	100	102	105	107	109	112	114	117	120	122	125	128	131	134	137	140	143	146	149	153	156

To change the base year of the SPCs from 2017 to any other year, the prices should be adjusted using the U.S. gross domestic product (GDP) deflator or seasonally adjusted U.S. consumer price index (CPI), starting in 2022. The Climate Change Group will update the SPCs once a year at the beginning of each fiscal year.

SPCs are subject to periodical review and revision may be undertaken when new evidence warrants such revision to the SPC values in the guidance note.

This note recommends two price trajectories—one high and one low—rather than one because of significant uncertainties arising from the unpredictability of future socioeconomic and technological trends, as well as how mitigation costs are expected to be borne in different countries, which depends on the country context as well as the willingness of the international community to transfer funds. With respect to country differentiation, the [Report of the High-Level Commission on Carbon Prices](#) acknowledged that the adoption of the same SPCs in every country would be optimal and equitable only if mitigation measures are implemented in order of increasing cost for cost efficiency *and* if unlimited resource transfers are made across countries to correct any inequitable distribution of the mitigation efforts. Absent such unlimited international transfers, the High-Level Commission recommended that SCCs be established at the national rather than the global level. These country-specific price levels and trajectories are influenced by a multitude of factors, including income level, poverty rates, economic structure, reliance on fossil fuels, potential for renewable energy sources (e.g., hydropower, wind, solar, and geothermal), and the ability of the government to support the transition and help the poor and vulnerable cope with the ensuing costs. If a country has determined an SPC trajectory consistent with its own decarbonization pathway, it is advised that teams report the outcomes using the country-specific

figures in addition to the analysis results utilizing the World Bank's recommended values depicted in Figure 1.

4. Overview of calculation and analysis

The objective of applying SPCs is to evaluate if the dollar benefits of the project are likely to outweigh the dollar costs, after valuing the carbon externalities attributed to the project over the economic life of each financed activity.

To account for carbon externalities in the economic analysis, the SPC (in U.S. dollars per tCO₂e) in each year is multiplied by the estimated annual absolute GHG emissions over the asset's economic life. The SPCs can be used in either CBA or CEA/LCA. In general, the task team will calculate the relevant indicators—such as the NPV and economic rates of return (ERR) in CBA and the PV in CEA/LCA—in the economic analysis.

- (1) without the SPCs,
- (2) with the low SPC trajectory, and
- (3) with the high SPC trajectory.
- (4) where CBA is carried out and the NPV is found to be positive at the low SPC trajectory but negative at the high SPC trajectory, the team should compute the trajectory for the switching values of SPC at which the NPV becomes zero.¹²

The interpretation of these results is presented in the project document along with the guidelines provided in the “Analysis” and “Reporting” sections below

The rest of section 4 is structured around the type of economic analysis. CBA is covered in section 4.1. LCA is described in section 4.2. Annex 2 presents a glossary of terms for different approaches used in this document.

Box 1. Defining project and baseline scenarios in the context of GHG accounting and economic analysis

Project scenario

The investment project is implemented with the WB's financial support.

Baseline scenario

The baseline scenario is the most feasible and realistic alternative scenario¹³ to an investment project, in many cases an alternative that can provide the same product or service within the same timeframe.¹⁴ It is important to note that a baseline is not necessarily a static single alternative investment project. The economic concept is to identify the mix of technological solutions or practices that would deliver

¹² The switching value is the maximum SPC below or above which the proposed project remains or becomes beneficial (e.g., NPV is positive); see details in the respective sections.

¹³ Referred to as the “next best alternative” in *Investment Project Finance Economic Analysis Guidance Note*.

¹⁴ For example, the World Bank Group's [Energy Sector Directions Paper](#) endorsed by its Board of Executive Directors in 2013 specifies that the baseline scenario for calculating relative GHG emissions should be drawn from alternatives to the project that deliver the same level of service within the same timeframe. By contrast, the traditional economic analysis of investment project financing is designed to assess the development impact, and in some cases such impact may be measured against a “no-project” scenario whereby no alternatives are implemented and the needs that would have been met by the WB project remain unmet.

the same service at the lowest possible cost, such as by changing the operation of or refurbishing existing assets, installing new assets, or doing both, subject to certain limits and conditions as appropriate.

How to select an appropriate baseline **in the context of GHG accounting** is normally explained in the Bank's GHG accounting methodologies and further under section 3.3 of the [*International Financial Institutions Guideline for a Harmonised Approach to Greenhouse Gas Accounting*](#) (IFI Guideline herein after). The IFI Guideline, breaks down options into three broad categories of baselines:

- (1) A **forward-looking baseline** captures recent advances in technology or management. For example, the baseline scenario may assume falling costs of energy storage or more effective demand management—such as a large reduction in the aggregate vehicle parking space available in the city center or expansion of congestion charging—and the project is compared against such a scenario.
- (2) The GHG emissions that represent the **average performance in the market, or the system**. It assumes that alternative providers of service will use available technologies, commonly adopted practices, or their equivalent.
- (3) A baseline based on **the existing actual emissions (without-project scenario, continuation of the current situation /status quo)**. The assumption embedded in this approach is that in the absence of the investment project, the service would be provided by the existing assets included in the assessment boundary at the historical level of GHG intensity (tCO₂e/unit of product or service) or the current practice (e.g., in the treatment of land or vegetation) would continue. This approach applies to, for example, energy efficiency investment projects in existing facilities with no capacity expansion or life extension, or to investment projects in existing land use activities. In these cases, existing demand will continue to be met even if doing nothing is the baseline, although at a higher cost and typically with higher emissions. However, a limitation of this approach becomes apparent when dealing with capacity expansion or life extension or both, as the historical GHG intensity neither reflects future shifts nor accounts for economy-wide changes (triggered by new regulations or technological advancements, for instance) that might lower the GHG intensity over the remaining lifetime of the asset.

For certain projects with high GHG emissions, the baseline scenario may be a function of the SPC. For example, if natural gas is displacing inexpensive coal for electricity or heat generation, coal-based energy generation may be the baseline without an SPC. However, as the price of carbon rises above a certain level, the baseline could shift to renewable sources such as solar or wind power, augmented by energy storage, with occasional reliance on natural gas or diesel to ensure reliability and flexibility. Thus, in scenarios where carbon pricing is a variable factor, task team should be careful about maintaining a coal-based energy generation as fixed baseline.

Where doing nothing and not meeting the demand met by the project is deemed to be the baseline, the task team should be careful to avoid the risk of overstating the benefits of the project. For example, if the cost of unmet demand is assumed to be high and captured as a benefit for the project—which is meeting otherwise suppressed demand—even a sub-optimal and otherwise uneconomic project can be made to have a high NPV and appear economic.

The baseline should be calculated for the same number of years as the economic life of the project asset responsible for GHG emissions. If the alternative asset has a shorter life, then another asset must replace it at the end of its life. If the cumulative number of life years of the assets in the baseline does not match the economic life of the project asset and exceeds it, one approach to approximating the corresponding costs with the same number of years as the project is to subtract the residual value of the assets in the baseline from the costs.

4.1. Applying SPCs in cost-benefit analysis (CBA)

In CBA, the project is considered economic if its NPV is positive—that is, the benefits exceed the costs—and the NPV is higher than those of viable alternatives. If the NPV is negative, it would likely be preferable not to implement the project. Where possible and feasible, the task team should look for an alternative with a positive NPV and explore if such an alternative could serve as the basis for reformulating the project. If the proposed project's NPV is positive but there is another economically viable project with a higher NPV, absent material barriers to its adoption, the latter should be chosen.

On the cost side, the economic analysis of the project without the application of SPCs includes (non-recurring) capital costs, (recurring) operating and maintenance costs, and any costs of externalities other than those associated with emissions of GHGs such as health damages from air pollution. These costs are incurred over the World Bank project's economic lifetime. To compute the net present value (NPV), discounted aggregate costs are subtracted from discounted benefits. When applying SPCs, the task team repeats the calculations, incorporating SPCs from the two price trajectories. Absolute emissions are calculated and multiplied by the SPCs, and the resulting costs are added to all other costs. If the emissions are negative, as in the case of afforestation, the cost of GHG emissions would be negative, effectively increasing the benefit stream.

Information on costs is usually more readily available than that on benefits, posing a challenge to the monetization of benefits. Benefits represent the aggregation of individual utility functions across time for the service provided by the project. In estimating benefits, it is important not to confuse revenues with benefits. Benefits are always higher than revenues (quantity multiplied by prices) and include consumer surplus and possibly other factors not captured in the analysis.

In many cases, information is not available to quantify the utility¹⁵ enjoyed by different beneficiaries at different points in time. As a result, broadly, there are two distinct approaches to CBA: (1) one using estimates of utility, and (2) the other using, in the place of utility for benefits, estimates of avoided costs of the most economic alternative to the project ("next best alternative" in [Investment Project Finance Economic Analysis Guidance Note](#)) if there is no obvious proxy for the aggregate utility enjoyed by consumers or beneficiaries of the service provided by the proposed project. These two approaches do not necessarily deliver the same NPVs but they do deliver the same ranking of options and the same selection of the most economic option, provided the same options are being compared.

¹⁵ In this section, we assume a cardinal utility function, which can be measured numerically. The main conclusions will also be valid if ordinal utility—which can be ranked but not measured numerically—is used. The difficulty in estimating cardinal utility is one of the reasons it is more common to estimate benefits based on avoided costs in practical applications of CBA.

One important difference is that if utility is estimated, it is not necessary for the options to be delivering the same level of service because benefits enjoyed by consumers at different levels of service are captured. When CBA using estimates of avoided costs is conducted, however, if the options examined do not provide the same levels of service, the results would not be meaningful. To identify the appropriate avoided costs, the task team should either (1) choose an option delivering the same level of service, or (2) if some options do not deliver the same level of service, add the cost of not meeting demand¹⁶ as a result of such non-delivery.

In determining whether a project is economic, it is insufficient to merely demonstrate a positive net present value (NPV). A cost-benefit analysis (CBA) that employs estimated utility must also compare the proposed project to the NPV of at least one alternative—the next best option. As in all other decision-making processes, the set of analyzed options should be realistically defined representing candidate technologies/measures and practices with the highest likelihood of best performance. Identification of alternatives for analysis, though potentially demanding, can be guided by initial economic assessments and technical due diligence. If, for example, the task team has identified several options, which have the best chances of being economic for the delivery of the service envisaged, and yet finds the NPV of every one of them to be negative, the team should also consider the option of not providing the service. For CBA using estimates of utility, the option of not providing the service has zero costs and zero utility. For CBA using avoided costs, however, the option of not providing the service may have a positive cost e.g., cost associated with unmet demand.

Calculation

The equation for calculating the NPV¹⁷ in the economic analysis of the project is

$$NPV = \sum_{t=1}^T \frac{Benefits_t - (capital\ and\ operational\ costs)_t - (SPC_t * absolute\ GHG\ emissions_t + other\ externalities)}{(1 + discount\ rate)^t}$$

----- Equation (1)

where time **t** runs from year 1 to the terminal year T, the latter of which typically represents the last year of the life of the project asset;

benefits represent the utility enjoyed by the users of the service provided, or alternatively avoided costs;

capital costs are non-recurring fixed costs.

operational costs are recurring fixed and variable operational and maintenance costs;

SPC_t: The value increases each year in real terms over the economic lifetime of the project, consistently with the pathways detailed in Figure 1. If a borrowing country has a domestic explicit carbon price or tax or receives carbon payments/credits for the emissions reduction delivered by a project and it has been

¹⁶ Meeting demand with substandard service quality is equivalent for the purpose of this note. Examples include unreliable supply of goods or services and goods not meeting the minimum specifications set for satisfactory performance.

¹⁷ It is important to note that it would be incorrect to use the difference in the emissions costs in this approach to CBA. Where utility is used to estimate benefits, negative net (relative) emissions may not be used to increase benefits of the proposed project in the economic analysis of the project

already included in the financial projections, then the carbon price or tax should be subtracted from the recommended SPCs in the economic analysis, so that the carbon externality is not accounted for twice.¹⁸

GHG emissions are the absolute emissions that occur during the implementation and operation of the WB project and can be positive (e.g., transport corridor project) or negative (e.g., afforestation);

other externalities capture harm from local air pollution, water contamination, congestion, and other cost of externalities associated with the activity in question; and

the **discount rate** is the social discount rate linked to the long-term growth prospects of the country where the project will be implemented as explained in [Discounting Costs and Benefits in Economic Analysis of World Bank Projects](#)

Analysis in which utility functions can be estimated.

If there are reasonable proxies to estimate aggregation of utility functions, SPCs affect the cost side of equation 1.¹⁹ A project that has a negative NPV without accounting for GHG emissions, cannot increase its own NPV unless absolute emissions are negative, such as a reforestation project. For all operations with positive absolute emissions, the NPV will decline with rising SPC.

If the NPV is positive, it does not automatically follow that the proposed project should be pursued. It should be compared to the most likely economically viable alternative (next best alternative) and the one with a higher NPV should be chosen. If the NPV of only the project is computed, then the next best alternative is implicitly assumed to be doing nothing and not providing the service supplied by the project, setting the NPV of the second best alternative equal to zero. However, without further analysis, it does not follow that doing nothing would be the next best alternative to the proposed project.

Analysis in which avoided costs are used to represent benefits

Here, the uncertainties associated with the estimation of the aggregate utility are considered substantial, although, some estimation may be needed even when avoided costs are used for benefits. A project delivering multiple goods or services may face even more information constraints. The avoided costs should be the costs of the next best alternative, which may not be the most viable alternative project selected as the baseline in the absence of SPCs, especially if the NPV of the baseline is negative.

Task teams conducting economic analysis with SPCs using the difference between the avoided costs corresponding to the next best alternative and the project costs should be careful to capture all avoided costs, and not just avoided emissions costs. This procedure is equivalent to subtracting all the costs of the next best alternative, inclusive of its emissions costs, from all the costs of the project, again inclusive of its emissions costs. The cost calculations are identical to those in LCA. In this approach to CBA, the difference between two sets of costs calculated in LCA is taken.

¹⁸ That is to say, the country's carbon price has no impact on the economic analysis using the World Bank's SPCs.

¹⁹ If a mechanism such as carbon taxes has been used to (partially or fully) internalize the economic costs of emissions in market prices, the actual carbon price determined via that mechanism will affect the market equilibrium and the utility side of the equation as well. However, there is no guarantee that the actual carbon price arising from a local and often sector-specific regulatory intervention will be numerically close to global and economy-wide marginal abatement costs, which are the basis for setting the SPCs in this document.

Analysis with no, low, and high SPC

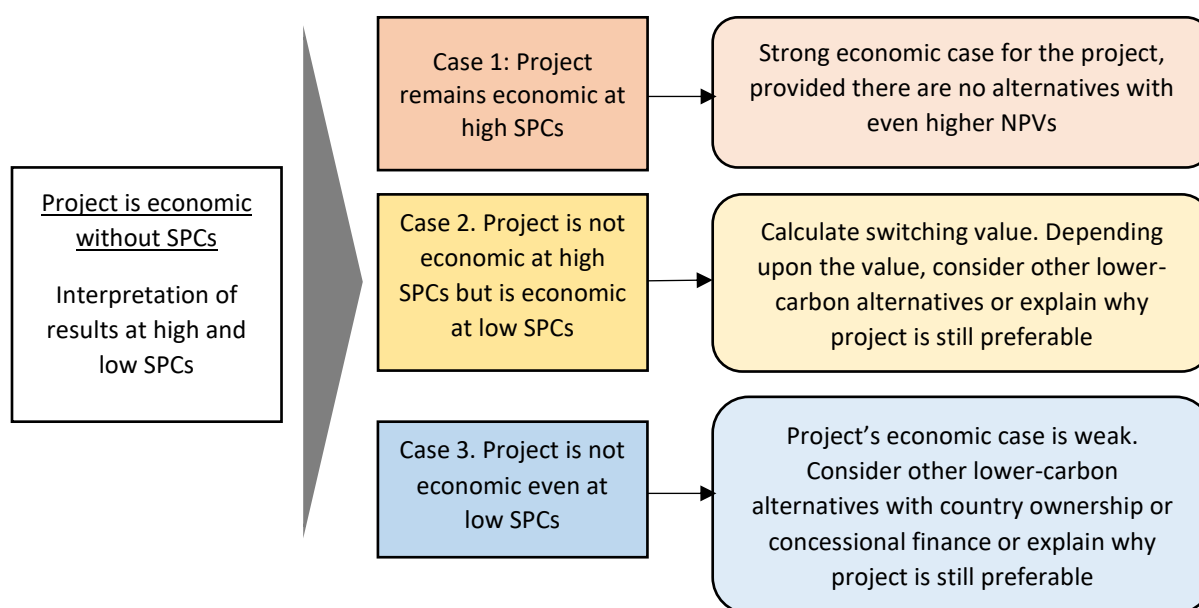
The task team analyzes the results of CBA with no, low, and high SPC—as well as the switching value where applicable—to evaluate if the WB project is economically beneficial. There are two possible results, which are mutually exclusive:

- (1) Projects that are beneficial without an SPC.
- (2) Projects that are beneficial only with an SPC.

Projects that are economic before accounting for GHG emissions

For projects for which the NPV is positive when GHG emissions are not accounted for, there are three possibilities, as shown in Figure 1. First, if the project has zero or negative absolute emissions, incorporating SPCs can only make the project even more economic by raising the NPV. If, however, the project has positive absolute GHG emissions, the analysis becomes more complex. The task team should compare the project's NPV with those of the next best alternative as well as non-provision of service to rank the three options. Where avoided costs are used, they may shift with increasing SPCs.

Figure 1 Interpretation of results of CBA for a project that is beneficial without SPCs



Note: “Economic” in this figure means that the NPV is positive and is the highest of the NPVs of the options examined, including the next best alternative.

- Case 1, which represents the most robust economic case for the project, is that the NPV is so strongly positive that even after adding the cost of emissions at high SPCs, the NPV remains positive and is the highest NPV of at least three options: project, next best alternative, and non-provision of the service.
- Case 2 is one in which the NPV is positive at low SPCs and remains the highest NPV of the options considered but becomes negative or is no longer the highest among the options at high SPCs. In this case, the task team computes the trajectory of SPCs that delivers an NPV of zero, also referred to as switching prices of carbon. The trajectory corresponding to such switching values will lie between the high- and low-price trajectories. The switching value can be calculated by finding the SPC for the first

year of calculation (and increasing at a constant rate of 2.25% throughout) that changes the sign of the NPV. If the switching value is low, then the team may want to re-examine the design of the project and explore other possible lower-carbon options for the WB project.²⁰ In addition, at the switching-value trajectory of SPCs, the task team should check if slightly decreasing SPCs from the switching values maintains the highest NPV ranking of the proposed project. If the initial project design is still preferred to other such options, the team can provide more information to explain why the project may still be desirable and describe constraints to other options or other non-monetized benefits of the preferred project choice.

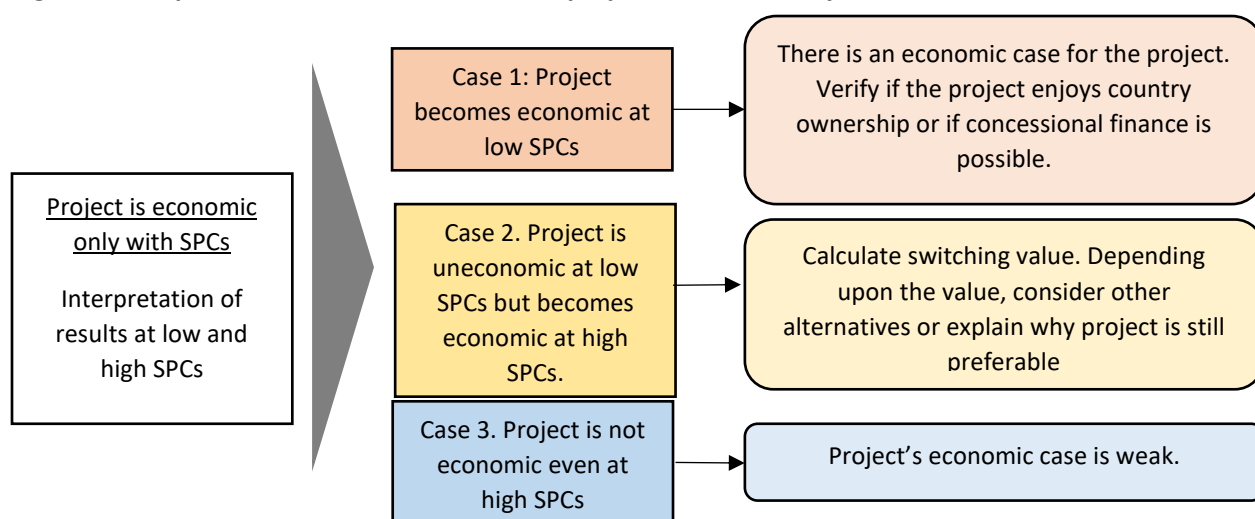
- Case 3 is one in which the NPV is negative, or positive but no longer the highest of the options considered even at low SPCs. In this case, the project's economic case is weak because even the low SPC trajectory threatens the project's overall economic value. Consideration may be given to exploring whether lower-carbon options can have country ownership (for example, through inclusion in a country's Nationally Determined Contribution), or whether incremental costs can be covered by concessional finance. If the team nevertheless decides to retain the initial project design, it can provide more information to explain the decision, such as constraints to selecting lower-carbon alternatives and non-monetized benefits of the selected project.

In cases 2 and 3, other options should be explored to achieve similar outcomes with lower GHG emissions.

Projects that are economic only with the inclusion of GHG emissions and SPCs

This category covers some projects with low or negative emissions. The NPV of a project that is negative in the absence of SPCs may turn positive once SPCs are captured. How to interpret the results with low and high SPCs is described in Figure 2.

Figure 2 Interpretation of results of CBA for a project beneficial only with SPC



Note: "Economic" in this figure means that the NPV is positive and is the highest of the NPVs of the options examined, including the next best alternative.

²⁰ However, projects with low switching values are appropriate in countries with lower capacity to reduce emissions (because of lower GDP per capita, low endowment in renewable energy potential, high poverty, highly carbon-intensive economic structure).

- Case 1, which represents the most robust economic case for the project, is that the inclusion of low SPCs is sufficient to turn the NPV positive and the highest NPV among at least three options: project, baseline, and non-provision of the service considered.
- Case 2 is one in which the NPV is negative at low SPCs but becomes not only positive but the highest NPV when the high SPC trajectory is used. In this case, the task team computes the trajectory of SPCs that delivers an NPV of zero, switching prices of carbon. The trajectory corresponding to such switching values will lie between the low- and high-price trajectories. The switching value can be calculated by finding the SPC for the first year of calculation (and increasing at a constant rate of 2.25% throughout) that changes the sign of the NPV. In addition, at the switching-value trajectory of SPCs, the task team should check if slightly increasing SPCs from the switching values makes the NPV of the project the highest among the three (or more) options being examined. If the switching value is high, then the team may want to re-examine the design of the project and explore other possible options for the WB project.²⁰ If the initial project design is still preferred to other such options, the team can provide more information to explain why the project may still be desirable and describe other non-monetized benefits of the preferred project choice.
- Case 3 is one in which the NPV remains negative even at high SPCs. In this case, the project's economic case is weak since even the high SPC trajectory does not deliver enough benefits to offset the costs. Consideration may thus be given to exploring other low-carbon options that are less costly and whether incremental costs can be covered by concessional finance. If the team nevertheless decides to retain the initial project design, it can provide more information to explain the decision, such as why the costly option enjoys country ownership, constraints to selecting less costly, low-carbon alternatives, and non-monetized benefits of the selected project.

Reporting

Concept stage: As mentioned above, analysis with SPCs is part of the economic analysis of investment project financing, for which the *Investment Project Finance Economic Analysis Guidance Note* provides more detailed guidance. In the case of CBA with multiple alternatives, the task team should identify at the concept stage multiple project alternatives that will be considered in the economic analysis with and without SPCs.

Appraisal stage: Task teams of applicable projects should present the economic analysis in the project appraisal document (PAD) to allow management and the Board to take this information into consideration. This SPC guidance note recommends that the PAD state the results of the economic analysis with high and low SPC values, and if relevant, provide the switching value.

In the case of a CBA with multiple alternatives, the NPVs for all assessed projects with no, low, and high SPCs should be presented. Additional information from qualitative assessment should be provided if relevant (e.g., on the constraints on other project options, reasons for selecting the project even if it is not least cost).

The team needs to report the NPV of the project for the different SPC values. If the NPV is negative or if there are other viable alternatives with higher NPVs as SPCs rise, additional explanations on the desirability of the project should be provided where applicable.

In addition, task teams should, before appraisal, report in the Appraisal Data Sheet under Institutional Data (can be found under Roadmap in the Operations Portal) whether they have used SPCs in the economic analysis (yes, no, or not applicable).

4.2. Applying SPC in cost-effectiveness or least-cost analysis

CEA and LCA are distinguished from CBA in not capturing benefits in the calculations (although, as explained above, the use of avoided costs to substitute benefits in CBA may be considered a variation of LCA). Instead, the costs of different feasible and realistic project options are compared on a discounted cash flow basis to rank the options in order of increasing PVs; only costs—capital and operating costs, and negative externalities, among which are GHG emissions—are valued in CEA/LCA. All options identified should provide the equivalent level of service within the same timeframe because otherwise costs can always be reduced by delivering fewer goods or services, invalidating the analysis, or else if one or more alternatives deliver lower levels of service or no service, the cost of unmet demand should be used for the service delivery gap at each point in time.²¹ The timeframe requirement has two dimensions. The first is the start of the delivery of service. The second is the duration of such delivery. Therefore, waiting longer to start delivery in the hope that the costs of an alternative will decline to make it the least cost, or comparing a costlier asset with a 30-year life with a lower-cost with a 15-year life, would not be appropriate.

As an illustrative example, alternatives to a thermal power generation project may comprise variable renewable energy with energy storage, back-up power generation, and other complementary measures to provide the same degree of service reliability. Further, if the life of the variable renewable energy asset is 15 years and the life of the thermal plant is 30 years, the former has to be replaced by a new asset halfway through the project cycle for comparison.

If a project has no GHG emissions and is already the lowest-cost option, then applying SPCs will not change the conclusion unless there is an alternative with negative absolute emissions. As with CBA, CEA and LCA are informative when at least one option being considered has positive absolute emissions. In that case, as SPCs are increased, an option with the lowest GHG emissions will eventually have the lowest PV. If the proposed has the lowest PV without taking GHG emissions into account but an alternative becomes the least-cost option with rising SPVs, the task team needs to calculate the switching value of SPCs at which the PVs of the proposed project and the alternative with lower PVs when SPCs are applied. As before, the switching values are calculated by seeking an SPC in the first year that makes the two PVs equal and that follows the same percentage increase each year as in Figure 1.

Calculation

The Present Value (PV) of each option can be calculated in absolute terms following equation (3):

²¹ In the case of CEA, it is important to consider the costs of unmet demand in the numerator and the quantity of unmet demand in the denominator of the performance metric. Using CEA in cases where at least one alternative has a substandard level or quality of service can make the interpretation of results more complex.

$$PV = \sum_{t=1}^T \frac{(Capital+operational)costs_t + (SPC_t * absolute\ GHG\ emissions_t + other\ externalities)}{(1+discount\ rate)^t}$$

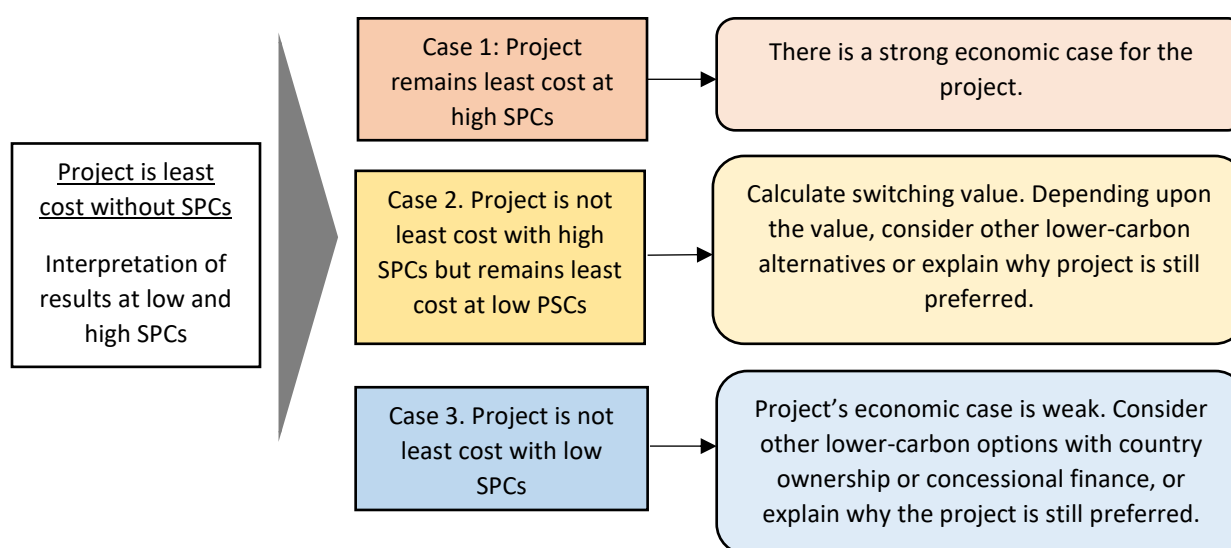
----- Equation (2)

Note: Definitions of parameters provided under equation 1 above, applies here also.

Analysis

For projects that have the lowest cost when compared to other feasible and realistic alternatives without the SPCs, the assessment of the results, with the high and low SPCs can be interpreted as shown in Figure 3. Whichever project has a lower PV would be the preferred option. If both emit GHGs, it is possible that the ranking of the corresponding PVs is reversed as the SPCs are raised from the low-price trajectory to the high-price trajectory.

Figure 3 Interpretation of CEA or LCA results for projects



There are three possible outcomes when applying CEA and LCA:

- Case 1 is one in which the considered project remains the least cost with the high SPCs. There is a strong economic case for the project since even the high SPCs do not change the ranking of different project options.
- Case 2 is one in which the considered project is no longer the least cost with the high SPCs but is the least cost with the low SPCs. In this case, the team needs to report the switching value. The switching value is the SPC at which an alternative to the proposed project has the same PV as the project and any higher SPC trajectory will reverse the ranking of PVs. The switching value can be calculated by finding the SPC for the first year of calculation (and increasing it at a constant rate of 2.25% throughout) that changes the selection of the least-cost project. If the proposed project is still preferred despite reversing the ranking based on respective PVs when SPCs are included, the task team can provide more information to explain the decision not to change the project design and describe the constraints in selecting lower-carbon project alternatives or other non-monetized benefits of the retained project.

- Case 3 is one in which the considered project is not the least cost with the low SPCs. The project's economic case is weak, as even a low SPC trajectory changes the ranking of different project options. Consideration may thus be given to exploring whether there may be country ownership of other lower-carbon project options (for example, through inclusion in a country's Nationally Determined Contribution), or whether incremental costs can be covered by concessional finance. If the task team decides to retain the initial project, the team can provide more information to explain such a decision or describe constraints in selecting lower-carbon project alternatives.

Reporting

Concept stage: [Investment Project Finance Economic Analysis Guidance Note](#) advises task teams to recommend the methodology to be used in economic analysis in the project concept note. This SPC Guidance Note explains how to capture SPCs in economic analysis where applicable. Thus, teams are to note that SPCs will be a part of the economic analysis.

In the case of CEA/LCA, teams should identify at the concept stage multiple project options that will be considered in the CEA/LCA with and without SPCs. Teams are encouraged to consider more systematically feasible and realistic lower-carbon alternatives that provide the same level of service within the same timeframe and compare them with the proposed project at low and high SPCs. If the PV of the preferred project choice is not the lowest after accounting for GHG emissions with SPCs, the team should explain why the project design has been retained. Such explanatory text is needed to prepare for questions likely to be raised by the Board.

Appraisal stage: Task teams of applicable projects should present the economic analysis in the PAD to allow management and the Board to take this information into consideration. This Guidance Note recommends that the PAD state the results of the economic analysis with high and low SPC values, and provide the switching values where applicable.

In CEA/LCA, the results for all assessed technology options with no, low, and high SPCs should be presented. Additional information from qualitative assessment should be provided if relevant (e.g., on the constraints in selecting other project options and reasons for the project even if it does not have the lowest cost).

In addition, task teams should, before appraisal, report in the Appraisal Data Sheet under Institutional Data (found under Roadmap in the Operations Portal) whether they have used SPCs in the economic analysis (yes, no, or not applicable).

5. Roles and responsibilities

Task teams can seek support from help desks that are available in each applicable Global Practice for the design of the baseline scenario and the review of the calculation of economic analysis with SPCs. GHG focal points in the Climate Change Group are available to support the GP GHG focal points and task teams in the analysis of projects that face exceptional circumstances.

Annex 1

Rationale for amendment to 2017 guidance note

The amendment aims to provide further clarity on the application of Shadow Prices of Carbon (SPCs) in the '2017 Guidance Note on Shadow Price of Carbon in Economic Analysis' across various types of economic analysis, including Cost-Benefit Analysis (CBA) and Life-Cycle Analysis (LCA). It offers a clearer structure for navigating the application of SPCs, supported by flowcharts, and illustrations of terminologies such as 'project scenarios' and 'baseline scenarios,' as well as streamlined sections. This amendment does not change the SPC values contained in the 2017 Guidance Note.

History of Revision

Rationale for revision from the 2014 guidance note

The guidance note was prepared in line with the recent developments in the climate change discussions after the release of 2014 “Social Value of Carbon in Project Appraisal Guidance Note”. The key developments and concerns included the following:

- **Paris agreement.** In 2015, the world came together and agreed to limit global warming to less than 2°C by 2100 and make best efforts to limit warming to 1.5°C. In the Climate Change Action Plan (CCAP) 2016-2020, the WB committed to support the achievement of these goals. Therefore, the recommended values are consistent with the core objective of the Paris Agreement of keeping temperature rise below 2°C and are based on the concept of marginal abatement cost only (while the 2014 guidance note also used the social cost of carbon, the value of climate change damages that can be avoided by reducing emissions).
- **Higher values.** Marginal abatement cost estimates are increasing over time (and will continue to do so) if short-term actions are not adequate for what the models consider cost-efficient. Hence, the values in the 2017 guidance note are higher than those in the 2014 guidance note.²²
- **Uncertainty.** There is significant uncertainty around the carbon value that is consistent with the Paris Agreement and this uncertainty increases over time. To reflect this uncertainty, the 2017 note recommended the use of low and high values (instead of a central estimate) and an approach to the assessment consistent with the presence of uncertainty.
- **Differentiated shadow prices of carbon.** Published in 2017, the report by the High-Level Commission acknowledges that uniform carbon prices are optimal only if unlimited resource transfers can be made across countries, so that countries with higher marginal abatement costs can finance abatement measures in countries with lower marginal abatement costs. Given the limitation on such transfers, the High-Level Commission report states that prices are better determined at the country level. Such country-specific price levels and pathways depend on many characteristics, such as income level, poverty incidence, economic structure and dependence to fossil fuels, potential of renewable

²² While not relevant here, the shadow price of carbon estimates have also increased due to updated analyses using recent evidence of climate change and variability impacts.

(hydropower, wind, and solar resources), and ability of the government to support the transition, especially for the poor and vulnerable. There are two (interlinked) reasons why lower-income countries may choose lower carbon prices than higher-income countries: (1) low-income countries tend to have less ambitious objectives for emission reductions; and (2) low-income countries tend to a lower carbon price to achieve a given level of emission reductions. Hence, the range between low and high values can be interpreted as the uncertainty combining technical uncertainty (what price level is appropriate to achieve a given level of GHG emissions reduction in a country?) and equity considerations (how fast should and could different countries reduce their emissions?).

- **At the request of task teams, the “Analysis”, “Reporting” and “Roles and Responsibilities” sections were added,** to clarify interpretation of, and operational context for the calculation of the SPC.

Annex 2 – Glossary of terms

Brownfield project vs greenfield project	Greenfield projects are those in new sites or in existing facilities where most of a plant and the equipment is new (e.g., road construction on previously unused land) and where, in the case of projects in existing facilities, all the critical items of equipment are decommissioned, or projects that primarily acquire and deploy new appliances or equipment. Brownfield projects modify existing facilities, equipment, appliances, systems, or processes (e.g., an upgrade of an existing road).
Cost-effectiveness analysis (CEA) and least-cost analysis (LCA)	In the case of cost-effectiveness or least-cost analysis, the costs of different realistically available project options that provide the same result (“same/equivalent level of service”) are compared. The CEA/LCA analysis typically compares multiple projects. CEA result: Costs per unit of benefit for <u>several projects</u> LCA result: Total costs of <u>several projects</u> ➔ Results available for no, low, and high SPCs (and if required, for switching value SPC)
Cost-benefit analysis (CBA)	CBA assesses the economic efficiency of projects through the systematic prediction and valuation (e.g., monetization) of costs and benefits. It considers the benefits accruing to and the costs incurred by all members of society.
Economically beneficial	Economically beneficial means that the project has a positive NPV

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